

ARAS with AI for two-wheelers

How Luna Systems aims to save lives on Indian roads?

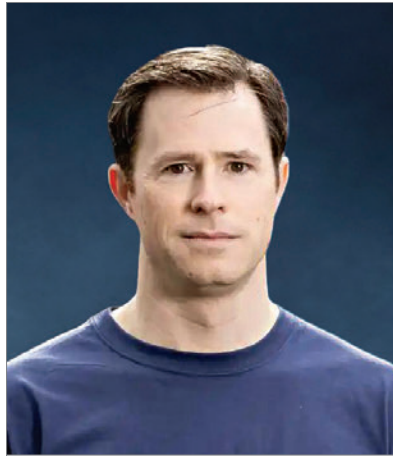
Jayesh Shinde | jayesh@digit.in

It started with a nagging question: Why do cars get all the fancy collision warnings and autopilot features, while two-wheelers – motorcycles and scooters – remain comparatively exposed, especially in a country like India where 44% of road fatalities involve two-wheelers?

That's the question that led me to a conversation with Andrew Fleury, CEO and Co-Founder of Luna Systems. For months, I'd been getting persistent emails about this little-known AI (computer vision) mobility safety company, culminating in a pitch about a demonstration at EICMA 2024, "the largest motorcycle show in the world." In an email interview, Fleury opened up about Luna's Advanced Rider Assistance Systems (ARAS) technology, the difficulties of scaling it in emerging markets like India, and how it might actually help reduce thousands of preventable road deaths every year.

"When we consider road safety on a global level, we primarily think of cars. However, according to the World Health Organisation, while cars represented 25% of global road fatalities in 2023, as much as 30% were two-wheeler related," Fleury explained, without mincing any words and getting straight to the point.

"This is in spite of a higher volume of cars versus two-wheelers being sold



Andrew Fleury
Co-Founder & CEO, Luna Systems

each year (80 million cars versus 56 million two-wheelers in 2023). At Luna Systems, we are monumentally inspired by the safety challenges faced by two-wheelers and also by our belief that everyone has the right to safe transport – regardless of the vehicle they choose," Fleury said further.

Reading that stark statistic drives home the urgency that Fleury and his team feel about bridging the massive gap between typical car-based ADAS solutions and the near absence of advanced safety technology for motorcycles and scooters – something that should be a prerequisite on Indian roads, especially.

Two-wheelers in India: A unique challenge

Few places could use this technology more than India. The country is rapidly becoming the epicentre for two-wheeler usage, with enormous sales

volumes and, unfortunately, staggering fatality numbers. "In 2023, 76,000 people died on Indian roads due to two-wheeler accidents – almost nine people every hour," Fleury pointed out, referencing official data. "This was 1,000 people more than the year previous and 7,000 more than 2021," he further pointed out.

One major barrier to ARAS adoption in India had been the historically high cost and complex integration. Cars get the full ADAS suite because each vehicle's price can absorb that extra cost. Motorcycles, on the other hand, often cater to a more price-conscious market. Another challenge is the chaotic nature of Indian roads. "From an ARAS perspective, India has indeed one of the most unique and therefore most challenging sets of road and traffic conditions," according to Fleury. Scooters, auto-rickshaws, pedestrians, stray animals – there's no shortage of unpredictable variables that can invade Indian roads, which are simply a whole another level of crazy than anywhere in the world.

How do you train an AI to cope with all of that craziness, I wonder? "We have an established partnership network in India, enabling us to continually build and expand our data sets and conduct real world testing and validation... We strongly believe that while speaking from a global perspective, there are commonalities in the nature of accident use cases, that the accident profile of every country is different," Fleury explained. That means collecting footage and telematics from Indian roads, labeling it meticulously, and refining the AI to handle bizarre situations – like an overloaded tractor crossing the wrong side of the median or a herd of goats occupying the highway.

